GITAM Deemed to be University

Aerospace Engineering | Course Code: AE301

Aerodynamics & Flight Mechanics - Comprehensive Syllabus Lecture Notes

Syllabus Focus: AE301 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Aerodynamics & Flight Mechanics. Includes det

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Fundamental Incompressible Potential Flow

" Syllabus Scope: Continuity and Euler equations, Velocity potential, Stream function, Elementary flows S

Unit 2: Incompressible Airfoil & Finite Wing Theory

" Syllabus Scope: Kutta condition, Thin airfoil theory for symmetric and cambered airfoils, Prandtl lifti

AE301: Aerodynamics & Flight Mechanics - Advanced Topics & Assessment

Academic Track: Aerospace Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Compressible Flow & Shock Waves

" Speed of sound, Mach number regimes, 1D isentropic flow through converging-diverging nozzles, Normal and obl

Unit 4: Aircraft Performance & Flight Stability

" Thrust required and power required curves, Rate of climb, Service ceiling, Static longitudinal stability, Ne

Unit 5: Hypersonic Aerodynamics & Viscous Effects

" Newtonian impact theory, Boundary layer transition, Aerodynamic heating, Swept wing benefits, Shock stall

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Derive the Kutta-Joukowski theorem L = rho * V_inf * Gamma for lift generation on a spinning cylinder
2. Using Thin Airfoil Theory, prove that the lift curve slope dC_L/dalpha equals 2*pi for a thin symmetri
3. Formulate the normal shock relations for static pressure ratio P2/P1 and downstream Mach number M2 in

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* John D. Anderson Jr. - Fundamentals of Aerodynamics McGraw-Hill 6th Ed
* E.L. Houghton, P.W. Carpenter - Aerodynamics for Engineering Students Butterworth-Heinemann

Authorized NoteVault Educational Document " Verified for Registered University Students